Supplement S1: Supplementary numerical results

Local Epochs, Aggregation and Variable Selection in Federated Coordinate-Descent Lasso

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The complete grid below accompanies the main manuscript. Every row summarises 200 generated datasets. Raw results, selected-fit costs, convergence diagnostics and scripts are available at https://github.com/KeivanBolouri/federated-lasso.

Table S1: Full simulation grid. Means over 200 replicates; Monte Carlo standard errors for F_1 in parentheses. Fit rounds and CD epochs include all five candidate fits for FedAvg-P. Upload includes their coefficient vectors and the validation scalars (2.81 KiB for ST; 0.35 KiB for P). Local penalty-search computation and all downloads are excluded. ADMM-CV is computed centrally using the equivalent pooled objective, so no federated cost is assigned. Other metrics and selected-fit costs are included in the accompanying CSV files.

Design	Method	E	Active	F_1 (MCSE)	$\|\hat{\beta} - \beta^{\star}\|_2$	Fit rounds	CD epochs	KiB
Independent	Pooled	—	120.6	0.405 (0.006)	1.93	—	—	—
	Pooled-1SE	—	48.1	0.715 (0.006)	2.29	—	—	—
	Local-only	—	107.3	0.395 (0.006)	2.85	0.0	—	0.0
	Local-only-1SE	—	39.0	0.582 (0.008)	3.31	0.0	—	0.0
	One-shot	—	194.0	0.264 (0.004)	2.91	1.0	—	14.1
	One-shot-1SE	—	53.5	0.547 (0.008)	3.67	1.0	—	14.1
	ADMM	—	41.6	0.743 (0.008)	2.59	58.2	—	818.2
	ADMM-CV	—	119.9	0.407 (0.006)	1.93	—	—	—
	ADMM-CV-1SE	—	48.7	0.715 (0.006)	2.30	—	—	—
	FedAvg	1	247.2	0.225 (0.004)	2.85	14.4	14.4	202.4
	FedAvg	2	223.8	0.245 (0.005)	2.87	9.1	18.2	127.8
	FedAvg	3	213.3	0.252 (0.005)	2.89	7.4	22.1	103.6
	FedAvg	5	205.2	0.258 (0.004)	2.90	5.9	29.4	82.6
	FedAvg	8	201.5	0.260 (0.004)	2.90	4.9	38.9	68.4
	FedAvg	10	200.2	0.261 (0.004)	2.90	4.5	45.0	63.3
	FedAvg	15	198.4	0.262 (0.004)	2.90	3.9	58.0	54.4
	FedAvg	20	197.7	0.262 (0.004)	2.90	3.5	69.8	49.1
	FedAvg-1SE	1	61.0	0.516 (0.008)	3.66	7.5	7.5	105.6
	FedAvg-1SE	2	54.9	0.543 (0.008)	3.67	5.0	9.9	70.0
	FedAvg-1SE	3	54.0	0.546 (0.008)	3.67	4.1	12.2	57.1
	FedAvg-1SE	5	53.6	0.547 (0.008)	3.67	3.2	15.9	44.6
	FedAvg-1SE	8	53.6	0.547 (0.008)	3.67	2.6	20.6	36.3
	FedAvg-1SE	10	53.6	0.547 (0.008)	3.67	2.3	22.8	32.1
	FedAvg-1SE	15	53.5	0.547 (0.008)	3.67	2.0	30.7	28.8
	FedAvg-1SE	20	53.5	0.547 (0.008)	3.67	2.0	40.2	28.3
	FedAvg-ST	1	43.2	0.667 (0.007)	3.24	14.4	14.4	205.2
	FedAvg-ST	2	43.5	0.664 (0.007)	3.26	9.1	18.2	130.6
	FedAvg-ST	3	44.2	0.657 (0.007)	3.27	7.4	22.1	106.4
	FedAvg-ST	5	44.6	0.654 (0.007)	3.28	5.9	29.4	85.4
	FedAvg-ST	8	44.7	0.652 (0.007)	3.29	4.9	38.9	71.2
	FedAvg-ST	10	44.7	0.651 (0.007)	3.29	4.5	45.0	66.1
	FedAvg-ST	15	44.8	0.650 (0.007)	3.29	3.9	58.0	57.2
	FedAvg-ST	20	44.9	0.649 (0.007)	3.29	3.5	69.8	51.9
	FedAvg-ST-min	1	225.6	0.240 (0.004)	2.85	14.4	14.4	205.2
	FedAvg-ST-min	2	211.3	0.254 (0.005)	2.88	9.1	18.2	130.6
	FedAvg-ST-min	3	203.8	0.259 (0.004)	2.89	7.4	22.1	106.4
	FedAvg-ST-min	5	197.4	0.264 (0.004)	2.90	5.9	29.4	85.4
	FedAvg-ST-min	8	195.4	0.265 (0.004)	2.90	4.9	38.9	71.2
	FedAvg-ST-min	10	193.9	0.266 (0.004)	2.90	4.5	45.0	66.1
	FedAvg-ST-min	15	192.5	0.267 (0.004)	2.91	3.9	58.0	57.2

Design	Method	E	Active	F_1 (MCSE)	$\|\hat{\beta} - \beta^{\star}\|_2$	Fit rounds	CD epochs	KiB
	FedAvg-ST-min	20	192.1	0.267 (0.004)	2.91	3.5	69.8	51.9
	FedAvg-P	1	62.1	0.557 (0.008)	3.21	60.7	60.7	854.3
	FedAvg-P	2	52.4	0.613 (0.008)	3.22	38.8	77.5	545.3
	FedAvg-P	3	51.5	0.618 (0.008)	3.23	31.3	93.9	440.5
	FedAvg-P	5	50.2	0.626 (0.008)	3.25	25.2	126.2	355.1
	FedAvg-P	8	50.4	0.624 (0.008)	3.25	21.1	169.1	297.6
	FedAvg-P	10	50.6	0.623 (0.008)	3.25	19.7	197.3	277.9
	FedAvg-P	15	50.4	0.624 (0.008)	3.25	17.3	259.3	243.4
	FedAvg-P	20	50.5	0.623 (0.008)	3.25	15.8	316.6	223.0
Correlated	Pooled	—	120.8	0.398 (0.005)	2.07	—	—	—
	Pooled-1SE	—	51.5	0.673 (0.006)	2.42	—	—	—
	Local-only	—	102.3	0.397 (0.005)	2.99	0.0	—	0.0
	Local-only-1SE	—	41.4	0.552 (0.006)	3.38	0.0	—	0.0
	One-shot	—	190.2	0.261 (0.003)	3.03	1.0	—	14.1
	One-shot-1SE	—	58.5	0.514 (0.006)	3.70	1.0	—	14.1
	ADMM	—	45.3	0.697 (0.007)	2.64	91.8	—	1291.4
	ADMM-CV	—	118.7	0.406 (0.006)	2.08	—	—	—
	ADMM-CV-1SE	—	52.2	0.668 (0.006)	2.41	—	—	—
	FedAvg	1	250.2	0.215 (0.004)	2.97	16.2	16.2	228.2
	FedAvg	2	223.3	0.237 (0.004)	2.99	10.2	20.4	143.3
	FedAvg	3	211.5	0.246 (0.004)	3.00	8.1	24.4	114.3
	FedAvg	5	201.7	0.254 (0.004)	3.01	6.3	31.7	89.2
	FedAvg	8	197.1	0.257 (0.004)	3.02	5.2	41.5	72.9
	FedAvg	10	195.5	0.259 (0.004)	3.02	4.8	47.5	66.8
	FedAvg	15	193.8	0.260 (0.004)	3.02	4.0	60.7	56.9
	FedAvg	20	193.0	0.260 (0.004)	3.02	3.6	72.2	50.8
	FedAvg-1SE	1	70.7	0.467 (0.007)	3.71	9.6	9.6	135.1
	FedAvg-1SE	2	61.2	0.504 (0.006)	3.70	6.4	12.8	89.6
	FedAvg-1SE	3	59.7	0.509 (0.006)	3.70	5.1	15.3	71.5
	FedAvg-1SE	5	58.9	0.513 (0.006)	3.70	3.9	19.6	55.1
	FedAvg-1SE	8	58.6	0.513 (0.006)	3.70	3.1	25.0	44.0
	FedAvg-1SE	10	58.6	0.513 (0.006)	3.70	3.0	29.5	41.5
	FedAvg-1SE	15	58.5	0.513 (0.006)	3.70	2.4	35.2	33.0
	FedAvg-1SE	20	58.5	0.513 (0.006)	3.70	2.1	41.7	29.3
	FedAvg-ST	1	53.6	0.590 (0.007)	3.26	16.2	16.2	231.0
	FedAvg-ST	2	53.5	0.588 (0.007)	3.29	10.2	20.4	146.1
	FedAvg-ST	3	54.0	0.582 (0.007)	3.30	8.1	24.4	117.1
	FedAvg-ST	5	54.5	0.579 (0.007)	3.31	6.3	31.7	92.0
	FedAvg-ST	8	55.0	0.575 (0.007)	3.31	5.2	41.5	75.7
	FedAvg-ST	10	55.2	0.573 (0.007)	3.32	4.8	47.5	69.6
	FedAvg-ST	15	55.3	0.572 (0.007)	3.32	4.0	60.7	59.7
	FedAvg-ST	20	55.3	0.572 (0.007)	3.32	3.6	72.2	53.6
	FedAvg-ST-min	1	238.0	0.221 (0.003)	2.97	16.2	16.2	231.0
	FedAvg-ST-min	2	215.3	0.241 (0.004)	2.99	10.2	20.4	146.1
	FedAvg-ST-min	3	205.1	0.250 (0.004)	3.00	8.1	24.4	117.1
	FedAvg-ST-min	5	196.8	0.257 (0.004)	3.01	6.3	31.7	92.0
	FedAvg-ST-min	8	192.9	0.260 (0.004)	3.02	5.2	41.5	75.7
	FedAvg-ST-min	10	191.3	0.261 (0.004)	3.02	4.8	47.5	69.6
	FedAvg-ST-min	15	190.6	0.262 (0.004)	3.02	4.0	60.7	59.7
	FedAvg-ST-min	20	189.7	0.262 (0.003)	3.02	3.6	72.2	53.6
	FedAvg-P	1	70.2	0.503 (0.007)	3.26	71.5	71.5	1006.1
	FedAvg-P	2	60.9	0.554 (0.007)	3.27	44.5	89.0	625.9
	FedAvg-P	3	58.8	0.562 (0.007)	3.28	35.1	105.5	494.6
	FedAvg-P	5	59.1	0.559 (0.007)	3.29	27.3	136.5	384.2
	FedAvg-P	8	59.5	0.557 (0.007)	3.30	22.5	180.0	316.7
	FedAvg-P	10	59.7	0.556 (0.007)	3.30	20.7	206.8	291.1
	FedAvg-P	15	59.9	0.556 (0.007)	3.30	18.0	269.6	253.1
	FedAvg-P	20	59.7	0.556 (0.007)	3.30	16.5	329.5	232.0
Heterogeneous	Pooled	—	117.5	0.386 (0.006)	2.73	—	—	—
	Pooled-1SE	—	32.3	0.596 (0.009)	3.41	—	—	—
	Local-only	—	107.3	0.395 (0.006)	2.85	0.0	—	0.0
	Local-only-1SE	—	39.0	0.582 (0.008)	3.31	0.0	—	0.0
	One-shot	—	181.7	0.273 (0.004)	3.10	1.0	—	14.1
	One-shot-1SE	—	55.1	0.523 (0.007)	3.74	1.0	—	14.1
	ADMM	—	37.3	0.529 (0.011)	3.42	65.8	—	925.2
	ADMM-CV	—	119.1	0.381 (0.006)	2.73	—	—	—
	ADMM-CV-1SE	—	33.1	0.596 (0.009)	3.39	—	—	—
	FedAvg	1	249.5	0.218 (0.004)	3.06	14.1	14.1	197.6

Design	Method	E	Active	F_1 (MCSE)	$\|\hat{\beta} - \beta^{\star}\|_2$	Fit rounds	CD epochs	KiB
	FedAvg	2	221.1	0.243 (0.005)	3.07	9.1	18.2	128.3
	FedAvg	3	206.8	0.254 (0.005)	3.07	7.5	22.6	105.8
	FedAvg	5	195.7	0.264 (0.004)	3.09	6.3	31.3	88.0
	FedAvg	8	190.4	0.268 (0.004)	3.09	5.3	42.4	74.5
	FedAvg	10	188.6	0.269 (0.004)	3.09	4.9	49.0	69.0
	FedAvg	15	186.4	0.271 (0.004)	3.10	4.3	64.1	60.1
	FedAvg	20	185.3	0.271 (0.004)	3.10	3.9	77.2	54.3
	FedAvg-1SE	1	69.1	0.475 (0.008)	3.74	7.9	7.9	110.7
	FedAvg-1SE	2	59.2	0.509 (0.008)	3.74	5.6	11.2	79.0
	FedAvg-1SE	3	57.1	0.517 (0.008)	3.74	4.8	14.3	66.9
	FedAvg-1SE	5	55.8	0.521 (0.007)	3.74	3.9	19.3	54.2
	FedAvg-1SE	8	55.2	0.522 (0.007)	3.74	3.2	25.9	45.6
	FedAvg-1SE	10	55.1	0.523 (0.007)	3.74	2.9	29.1	41.0
	FedAvg-1SE	15	55.1	0.523 (0.007)	3.74	2.6	39.7	37.2
	FedAvg-1SE	20	55.1	0.523 (0.007)	3.74	2.4	47.7	33.5
	FedAvg-ST	1	23.2	0.636 (0.013)	3.76	14.1	14.1	200.5
	FedAvg-ST	2	23.8	0.638 (0.012)	3.76	9.1	18.2	131.1
	FedAvg-ST	3	23.8	0.641 (0.012)	3.76	7.5	22.6	108.6
	FedAvg-ST	5	23.9	0.638 (0.012)	3.77	6.3	31.3	90.8
	FedAvg-ST	8	24.0	0.638 (0.012)	3.78	5.3	42.4	77.3
	FedAvg-ST	10	24.0	0.636 (0.012)	3.78	4.9	49.0	71.8
	FedAvg-ST	15	24.2	0.634 (0.012)	3.78	4.3	64.1	62.9
	FedAvg-ST	20	24.2	0.635 (0.012)	3.78	3.9	77.2	57.1
	FedAvg-ST-min	1	233.1	0.232 (0.005)	3.06	14.1	14.1	200.5
	FedAvg-ST-min	2	210.4	0.252 (0.005)	3.07	9.1	18.2	131.1
	FedAvg-ST-min	3	198.0	0.261 (0.005)	3.08	7.5	22.6	108.6
	FedAvg-ST-min	5	187.3	0.272 (0.005)	3.09	6.3	31.3	90.8
	FedAvg-ST-min	8	183.9	0.274 (0.004)	3.09	5.3	42.4	77.3
	FedAvg-ST-min	10	182.9	0.275 (0.004)	3.10	4.9	49.0	71.8
	FedAvg-ST-min	15	181.1	0.277 (0.004)	3.10	4.3	64.1	62.9
	FedAvg-ST-min	20	179.8	0.277 (0.004)	3.10	3.9	77.2	57.1
	FedAvg-P	1	30.2	0.542 (0.014)	3.76	50.8	50.8	715.3
	FedAvg-P	2	26.6	0.585 (0.015)	3.75	34.3	68.6	482.8
	FedAvg-P	3	26.0	0.591 (0.015)	3.76	28.7	86.0	403.5
	FedAvg-P	5	26.1	0.590 (0.015)	3.77	23.7	118.7	334.1
	FedAvg-P	8	26.2	0.591 (0.015)	3.77	20.5	163.8	288.3
	FedAvg-P	10	26.0	0.589 (0.015)	3.77	19.1	190.9	268.9
	FedAvg-P	15	26.1	0.589 (0.015)	3.78	17.0	254.8	239.2
	FedAvg-P	20	26.1	0.588 (0.015)	3.78	15.6	312.4	220.0

Common total-upload cap

Table S2 uses the archived common-penalty experiment under one upload cap for both methods, including all learning-rate candidates. It is a sensitivity analysis added after the original runs. The largest recorded affordable checkpoint is chosen by cost alone: 100 rounds for CD-common and 20 for adapted FedDualAvg. The latter's three paths are all charged. Unequal actual spending reflects unused budget on the coarse checkpoint grid, not omitted candidate costs. No outcomes are interpolated and no new trajectories are fitted. The paired differences and Monte Carlo standard errors appear in `budget_total_cap_paired.csv` in the repository.

Table S2: Sensitivity analysis under a common total-upload cap of 1,440,120 bytes (1406.37 KiB) per method, design and E. The largest affordable recorded checkpoint is used: CD has $R = 100$ and DA has $R = 20$. Upload includes every candidate path and standalone setup and validation scalars. Means (Monte Carlo standard errors) over 50 paired replicates. The coarse checkpoint grid leaves unequal unused budgets; actual spending is not matched.

Design	Method	E	Total KiB	F_1	Objective gap	Test MSE
Independent	CD	1	1406.27	0.256 (0.013)	1.695 (0.051)	29.666 (0.564)
	CD	5	1406.27	0.295 (0.014)	1.614 (0.054)	30.314 (0.558)
	CD	20	1406.27	0.298 (0.014)	1.642 (0.057)	30.380 (0.559)
	DA	1	843.87	0.695 (0.018)	0.033 (0.004)	28.807 (0.630)
	DA	5	843.87	0.759 (0.016)	0.007 (0.001)	28.207 (0.654)
	DA	20	843.87	0.760 (0.015)	0.010 (0.002)	28.219 (0.651)
Correlated	CD	1	1406.27	0.235 (0.007)	1.665 (0.037)	28.990 (0.518)
	CD	5	1406.27	0.278 (0.008)	1.530 (0.034)	29.549 (0.511)
	CD	20	1406.27	0.282 (0.007)	1.548 (0.035)	29.588 (0.511)
	DA	1	843.87	0.612 (0.013)	0.122 (0.009)	28.890 (0.515)
	DA	5	843.87	0.742 (0.010)	0.005 (0.000)	27.578 (0.530)
	DA	20	843.87	0.749 (0.010)	0.010 (0.001)	27.641 (0.531)
Heterogeneous	CD	1	1406.27	0.234 (0.008)	3.122 (0.086)	44.771 (0.822)
	CD	5	1406.27	0.277 (0.007)	2.448 (0.066)	44.913 (0.804)
	CD	20	1406.27	0.285 (0.007)	2.475 (0.067)	44.997 (0.798)
	DA	1	843.87	0.477 (0.022)	0.119 (0.013)	44.066 (0.879)
	DA	5	843.87	0.565 (0.027)	0.069 (0.008)	43.198 (0.940)
	DA	20	843.87	0.573 (0.029)	0.093 (0.018)	43.245 (0.935)